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INVITED PAPER

Ultralow-Power Design in Near-Threshold Region

New basic logic that selects between possible output values using a sense amplifier may be able to dramatically improve ultralow-power system performance.

By Dejan Marković, *Member IEEE*, Cheng C. Wang, *Student Member IEEE*, Louis P. Alarcón, *Student Member IEEE*, Tsung-Te Liu, *Student Member IEEE*, and Jan M. Rabaey, *Fellow IEEE*

ABSTRACT | Operation in the subthreshold region most often is synonymous to minimum-energy operation. Yet, the penalty in performance is huge. In this paper, we explore how design in the moderate inversion region helps to recover some of that lost performance, while staying quite close to the minimum-energy point. An energy-delay modeling framework that extends over the weak, moderate, and strong inversion regions is developed. The impact of activity and design parameters such as supply voltage and transistor sizing on the energy and performance in this operational region is derived. The quantitative benefits of operating in near-threshold region are established using some simple examples. The paper shows that a 20% increase in energy from the minimum-energy point gives back ten times in performance. Based on these observations, a pass-transistor based logic family that excels in this operational region is introduced. The logic family operates most of its logic in the above-threshold mode (using low-threshold transistors), yet containing leakage to only those in subthreshold. Operation below minimum-energy point of CMOS is demonstrated. In leakage-dominated ultralow-power designs, time-multiplexing will be shown to yield not only area, but also energy reduction due to lower leakage. Finally, the paper demonstrates the use of ultralow-power design techniques in chip synthesis.

KEYWORDS | Circuit optimization; CMOS digital integrated circuits; curve fitting; design methodology; integrated circuit design; integrated circuit modeling; leakage currents; logic design; optimization methods; pipelines; sensitivity

I. INTRODUCTION

Traditionally, design optimization in the logic circuit community has always targeted the minimum-delay operational point (MDP). When energy efficiency became an issue, it was learned that backing off a couple of percentages from the minimum-delay point yielded a substantial amount of energy savings [1]. Yet, it has taken quite some time for designers to start exploring the overall energy-delay space. Over the past decade, it was shown that for a given design and technology, a pareto-optimal design curve can be derived as shown in Fig. 1, which uniquely defines the minimum energy needed for a given performance (and vice versa) over a set of design parameters such as the supply and threshold voltages, as well as transistors sizes, [2]–[6]. Yet the analysis tools used to derive this trade-off curve were primarily focused on the high-performance side, and the accuracy at the low-energy end was questionable at best.

With energy rapidly becoming the most compelling design concern, focus has shifted. Research over the past five years has demonstrated that the extreme opposite, that is the minimum-energy point (MEP) of the energy-delay curve, is attractive as well. With the emergence of sensor and biomedical applications that require ultralow energy, we have to adequately address design optimization near the minimum-energy point. This represents an important mind-shift: rather than starting out from a design optimized for maximum performance, the initial design point is now the minimum-energy one. It has long been established that for

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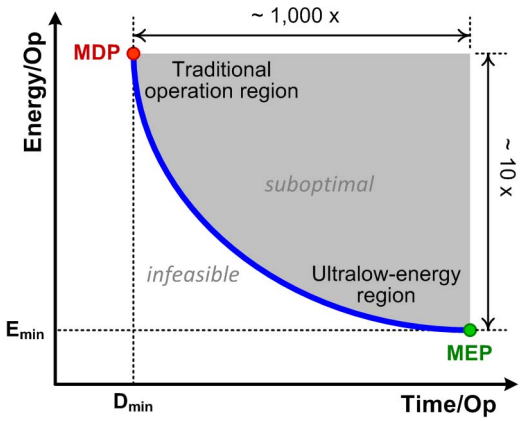

Fig. 1. ***Energy-delay trade-off in combinational logic. Traditional operation region is around minimum-delay point (MDP). Ultralow-energy region is around minimum-energy point (MEP).***

most logic families the minimum-energy point occurs in the subthreshold operational region of the MOS transistors [7]–[9], and that the value of the minimum energy is set by leakage. Based on these observations, subthreshold digital logic design has gained quite some attention over the past few years [9], [10]. These efforts have resulted in some remarkable results, and have demonstrated that minimum-energy design is indeed possible. The work of the Michigan group, for instance, has demonstrated that a programmable processor can be realized that consumes only 2.6 pJ per instruction (equivalent to 2.6 mW per GHz instruction), [10]. Some of the pitfalls of subthreshold design have been exposed as well. First, in contrast to operation in strong inversion, subthreshold logic has to be ratioed to be functional (as logical levels are set by balancing various leakage currents). Also, increased sensitivity to parameter variance is a concern as well.

As was the case in the maximum-performance scenario, the penalty for operating at the minimum-energy point is substantial. The delay at MEP is at least three orders of magnitude larger than at MDP, Fig. 1. Fortunately, as the optimum energy-delay curve is quite flat around MEP, a substantial amount of performance can be recovered by just backing off a bit. This unfortunately moves the operational point of the design in between the weak and moderate inversion regions of the MOS transistor, commonly called the near-threshold region. This operational region has traditionally not been very well covered by modeling, let alone design optimization tools. Hence determining the design parameter values that ensure minimum-energy operation while guaranteeing a required performance level is non-trivial.

In this paper, we present an environment that allows for accurate modeling of delay and energy in the near-threshold region as a function of the essential design parameters. In addition, it enables us to extend the sensitivity-based optimization strategy, presented in [1], [32] to the ultralow-power domain. Using this newly gained capability, it is now possible to weigh the trade-offs between transistor sizing and voltage scaling. Predominantly sizing-based optimization for high-performance will be replaced with voltage-based optimization for ultralow-energy designs. Also, the benefits of using various architectural techniques such as concurrency, pipelining, or time-multiplexing to reduce energy consumption can now be examined in earnest. Key optimization concepts derived from our methodology will be applied to practical designs within commercial chip synthesis flows. Finally, the methodology allows us to examine what features a logic circuit family should have to operate efficiently in the ultralow-energy regime. Based on this, we examine the effectiveness of a pass-transistor based family, called sense-amplifier-based pass-transistor logic (SAPTL) that is capable of operating below MEP of static CMOS logic. The use of architectural techniques for energy-delay optimizations examines simultaneous area and energy savings due to reduction in leakage energy, which is dominant around MEP.

The rest of the paper is structured as follows. First, a simple model for MOS transistor current as a function of voltages that fits the moderate and weak inversion regions is derived. Essential is that the model matches particularly well the behavior around the threshold voltage ($V_{GS} = V_T$) and in the off-mode ($V_{GS} = 0$). Based on this, gate delay and energy models that can be used in logic analysis tools based on logical effort [1], [11] are derived in Section II. An optimization framework built on sensitivity analysis in Section III is constructed and used to analyze some simple logic modules in Section IV. Section V discusses the impact of using leakage-insensitive logic families such as SAPTL to demonstrate operation below MEP of static CMOS. Another method to efficiently control the threshold via leakage reduction can be achieved by adjustment of the architectural parameters, as discussed in Section VI. The use of ultralow-power (ULP) design methods within chip synthesis environment is presented in Section VII. A design example is also provided to demonstrate superior energy and area efficiency that are verified by silicon measurements. Section VIII concludes the paper.

II. DEVICE AND CIRCUIT MODELS

This section presents a compact model for the MOS drain current, delay, and energy of an MOS gate operating in the moderate and weak inversion regions. The model is derived by curve-fitting Spectre simulation results for a 65-nm CMOS technology (considering both the high- and low-V_T process options). The fitting is performed such that the same set of technology parameters is used for both operational regions, while ensuring continuity for optimization purposes.

A. Current Model

The device current model must be accurate in the weak ($V_{GS} < V_T$) and moderate ($V_{GS} > V_T$) inversion regions to

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adequately model delay. In addition, accuracy is also essential at $V_{GS} = 0$ to ensure that leakage currents are modeled correctly. A convenient starting point is the onset of inversion ($V_{GS} = V_T$) where the current can be expressed as:

$$I_S = 2 \cdot n \cdot \mu \cdot C_{\text{ox}} \cdot \frac{W}{L} \cdot \phi_t^2. \tag{1}$$

The model in (1) is based on the EKV formulas [7], with the subthreshold slope n, mobility μ, oxide capacitance C_{ox}, and thermal voltage $\Phi_t = kT/q$ as parameters. The current in the vicinity of V_T can then be modeled as:

$$I_{DS} = \frac{I_S \cdot IC}{k_{\text{fit}}}. \tag{2}$$

Here, IC represents the *inversion coefficient*, and k_{fit} is a model-fitting parameter. The inversion coefficient expresses the degree of inversion of the transistor, and covers both the sub-V_T ($IC < 1$) and above-V_T ($IC > 1$) regions. While the introduction of the IC parameter leads to simple current expressions, the link to supply voltage is somewhat lost, as IC is a strongly non-linear function of V_{DD} as described in (3):

$$IC = \left(\ln\left(e^{\frac{(1+\sigma)\cdot V_{DD} - V_T}{2\cdot n\cdot \phi_t}} + 1\right)\right)^2, \text{ or}$$
$$V_{DD} = \frac{V_T + 2 \cdot n \cdot \phi_t \cdot \ln(e^{\sqrt{IC}} - 1)}{1 + \sigma} \tag{3}$$

in which σ represents the DIBL factor.

The leakage current at $V_{GS} = 0$ can be exressed using (4), based on the EKV formulas:

$$I_{\text{Leakage}} = I_s \cdot e^{\frac{\sigma \cdot V_{DD} - V_T}{n \cdot \phi_t}}. \tag{4}$$

Finally, we must ensure that models around threshold (2) and at the cutoff point (4) are based on the same set of technology parameters, which in our case will be accomplished by curve-fitting to transistor-level simulations. Generally, such curve-fitting approach makes it hard to predict scaling trends, but the presented methodology can be used to quickly estimate fitting parameters for any technology. The objective is, thus, to develop compact yet accurate models for design optimizations.

For a 65-nm CMOS technology, we sweep V_{DD} from 0.1 V to 0.6 V to extract model parameters for (2) and (4) as shown in Figs. 2 and 3. Two process options, high-V_T and low-V_T, are considered to derive general process parameters for the technology and to compare different device options for ultralow-power design. We restrict the model

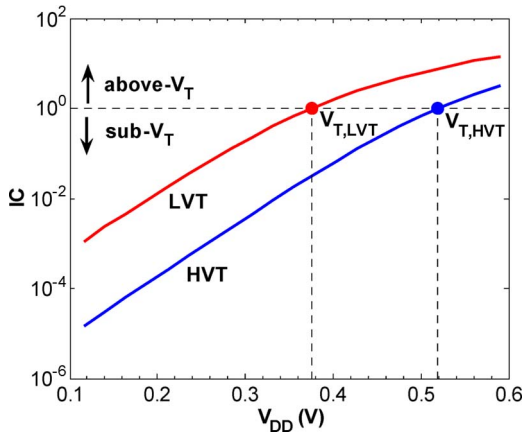

Fig. 2. ***Inversion coefficient for HVT and LVT devices for a 65 nm technology.***

so that $IC = 1$ corresponds to $V_{GS} = V_T$. This means that $n = (\sigma \cdot V_T)/(2 \cdot \phi_t \cdot \ln(e - 1))$, as derived from (3).

We performed a simultaneous fitting of the parameters to both low-V_T (LVT) and high-V_T (HVT) transistors, as shown in Fig. 3. Since a single set of fitting parameters is used for both types of transistors, the mean-squared-error increased from 0.5% to around 1.5%, but the model is still very accurate. This allows us to explore multi-V_T optimization in the energy-delay space. The current model will be used next as a baseline for the derivation of delay and energy models.

B. Delay Model

Based on the current model from the previous section, the model for delay analysis can be derived [12], [13]. Substituting (1) and (2) into the alpha-power law model for delay, the gate delay can be expressed as:

$$t_p = \frac{k_{tp} \cdot C_L \cdot V_{DD}}{2 \cdot n \cdot \mu \cdot C_{\text{ox}} \cdot \frac{W}{L} \cdot \phi_t^2} \cdot \frac{k_{\text{fit}}}{IC} \tag{5}$$

where k_{tp} is the delay-fitting parameter.

A restructuring of (5) is helpful to make the impact of gate sizing and operating voltage more explicit (as needed for design optimization). The capacitance C_L is the sum of the intrinsic capacitance of the driving stage and the load capacitance of the fanout gates. The W in the denominator of (5) stands for the width of the transistors in the driving stage. For path-delay analysis, we annotate the driver and load gates as stages i and $i + 1$, respectively. Thus, C_L is proportional to $C_{\text{ox}} \cdot L \cdot (\gamma_i \cdot W_i + W_{i+1})$, where $\gamma = C_{\text{parasitic}}/C_{\text{in,gate}}$ is the ratio of gate parasitic to input capacitance, L is the channel length, and C_{ox} is the oxide capacitance. In this notation, W in the denominator of (5)

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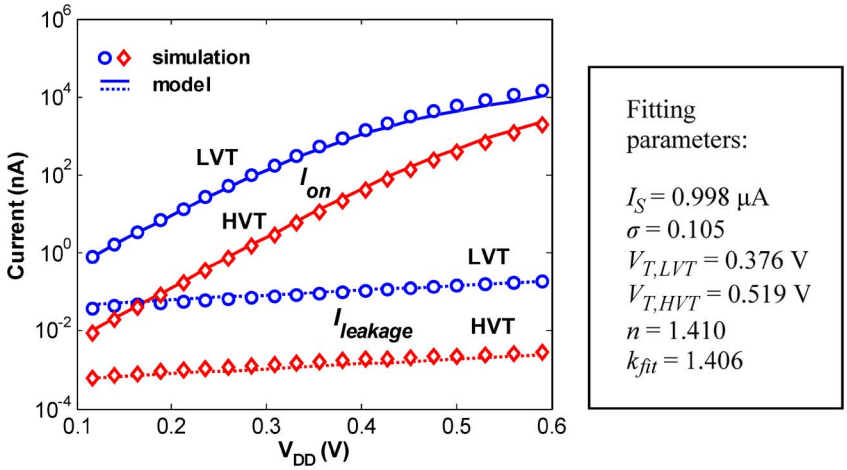

Fig. 3. ***Drain current model covering both high-V_T and low-V_T devices. The fit was done such that $IC = 1$ for $V_{DD} = V_T$. Fitting error (model vs. simulation): 1.5%.***

becomes W_i. Based on above observations, we can rewrite the propagation delay expression:

$$t_p = \frac{k_{tp} \cdot k_{\text{fit}} \cdot L^2}{2 \cdot n \cdot \mu \cdot \phi_t^2} \cdot \frac{V_{DD}}{IC} \cdot \frac{\gamma_i \cdot W_i + W_{i+1}}{W_i}. \tag{6}$$

The delay of a path D can now be derived as the sum of the gate delays:

$$\begin{aligned} D &= \frac{k_{tp} \cdot k_{\text{fit}} \cdot L^2}{2 \cdot n \cdot \mu \cdot \phi_t^2} \cdot \frac{V_{DD}}{IC} \cdot \sum_{i=1}^{N} \frac{\gamma_i \cdot W_i + W_{i+1}}{W_i} \\ &= K_d \cdot \frac{V_{DD}}{IC} \cdot D(W) \end{aligned} \tag{7}$$

where the first factor depends solely on technology and gate topology, the second upon supply and threshold voltage, and the third one on sizing. The path-delay is often expressed in terms of average gate delay t_p and the logic depth L_d, or as $D = t_p \cdot L_d$. Such formulation is convenient for fast cycle-time estimation at the microarchitectural level, but cannot be used for gate sizing of the underlying logic paths. The term $D(W)$ will allow us to perform gate sizing optimization.

C. Energy Model

The energy consumed by a gate is the sum of the active (or switching) energy, as well as the leakage energy consumed during the clock period (the clock period is assumed to be equal to D). For a single gate, this can be expressed as follows:

$$\begin{aligned} e_{\text{gate}} &= e_{sw} + e_{lk} \\ &= \alpha \cdot C_L \cdot V_{DD}^2 + D \cdot I_S \cdot e^{\frac{\sigma \cdot V_{DD} - V_T}{n \cdot \phi_T}} \cdot V_{DD}. \end{aligned} \tag{8}$$

Here, α represents the switching activity of the gate, which is the probability that the gate switches during the period D. It is worth observing that the load capacitance C_L is a function of the size of the gate and its fanout as previously discussed. I_S is only a function of gate size itself, while D is a function of the size ratios as in (7). These dependencies are stated more explicitly below:

$$\begin{aligned} e_{\text{gate,i}} = {} & \alpha_i \cdot C_{\text{ox}} \cdot L \cdot (\gamma_i \cdot W_i + W_{i+1}) \cdot V_{DD}^2 + K_d \\ & \cdot \frac{V_{DD}}{IC} \cdot D(W) \cdot \left(2 \cdot n \cdot \mu \cdot C_{\text{ox}} \cdot \frac{W_i}{L} \cdot \phi_t^2\right) \\ & \cdot e^{\frac{\sigma \cdot V_{DD} - V_T}{n \cdot \phi_t}} \cdot V_{DD}. \end{aligned} \tag{9}$$

The energy consumed over a complete path can be obtained through summation. To simplify the expression, we isolate factors dependent upon supply voltage and device sizes, and group technology constants into single parameters.

$$E = \left[K_{sw} \cdot \alpha \cdot \sum_{i=1}^{N} (\gamma \cdot W_i + W_{i+1}) + K_{lk} \cdot D(W) \cdot \sum_{i=1}^{N} W_i \cdot \frac{e^{\frac{\sigma \cdot V_{DD} - V_T}{n \cdot \phi_t}}}{IC} \right] \cdot V_{DD}^2 \tag{10}$$

or

$$E = \left[K_{sw} \cdot \alpha \cdot E_{sw}(W) + K_{lk} \cdot D(W) \cdot E_{lk}(W) \cdot \frac{e^{\frac{\sigma \cdot V_{DD} - V_T}{n \cdot \phi_t}}}{IC} \right] \cdot V_{DD}^2 \tag{11}$$

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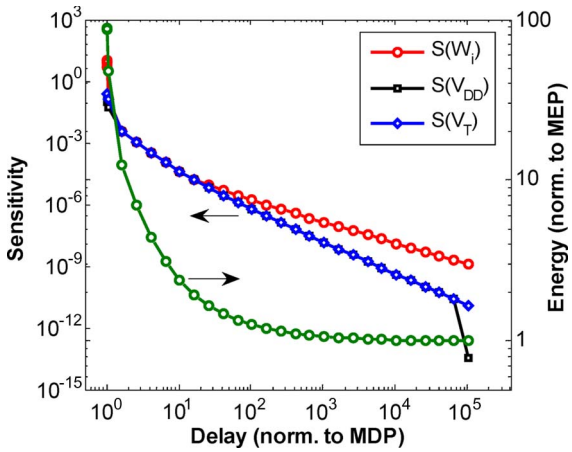

Fig. 4. ***Energy-delay sensitivity*** $S(x) = (\partial E/\partial x)/(\partial D/\partial x)$ ***to sizing*** (W_i)***, supply*** (V_{DD}) ***and threshold*** (V_T) ***voltage (left y-axis), and energy-delay trade-off (right y-axis) for a 32-bit carry look-ahead adder.***

In (10) and (11), α represents the activity factor of the datapath, or the average activity for all gates. K_{sw} and K_{lk} represent technology (and fitting) constants. Note that the energy-per-operation E_{op} is the path energy E divided by the activity factor α, $E_{\mathrm{op}} = E/\alpha$. As activity approaches zero, E_{op} would approach infinity. This may seem counterintuitive at first, but makes sense because no operation is performed at zero activity yet (leakage) energy is being dissipated. Separation of voltage- and size-dependent parameters in (11) will prove useful in the derivative analysis, discussed next.

III. SENSITIVITY ANALYSIS

In this section, we present a framework to analyze the impact of gate sizing, supply and threshold voltage on energy-delay trade-offs. The energy-delay trade-offs via voltage and gate sizing will be quantified using the concept of energy-delay sensitivity. The sensitivity to a parameter x represents a percent reduction in energy for a percent increase in delay, $S(x) = (\partial E/\partial x)/(\partial D/\partial x)$, [1], [31], [32]. Previous work [1], [32] has shown that sizing was the most effective around MDP. Here, the emphasis will be placed on the trade-offs around MEP. Let's examine the sensitivities of the optimization parameters along the optimal energy-delay (E-D) curve.

Fig. 4 shows simulated energy-delay sensitivity for an adder as well as optimal E-D trade-off when gate sizing, supply and threshold voltage are varied. Fig. 5 shows a closer look into areas around MDP [Fig. 5(a)] and MEP [Fig. 5(b)] to compare techniques for high-performance and low-energy design optimization. On the optimal E-D curve, the sensitivities of the active parameters are equal. Lower sensitivity represents more delay reduction for a fixed energy increase or less increase in energy for a fixed delay reduction. When the sensitivity to a parameter deviates from the lowest curve, such parameter has reached its constraint limit, and is no longer active to support further energy reduction. This is the case with V_T and sizing (W_i) at MEP [Fig. 5(b)], and V_T and V_{DD} at MDP [Fig. 5(a)]. As expected, near MEP, V_{DD} adjustment has the lowest sensitivity (it has least increase in energy for a given delay reduction), and thus the most effective parameter in delay reduction. Notice that we are looking at energy-delay sensitivity. Delay-energy sensitivity (as a measure of delay improvement for a given energy increase) to V_{DD} would be the highest, just like E-D sensitivity to sizing is the highest around MDP. As we traverse up the E-D curve, from Fig. 5(b) to Fig. 5(a), V_T also becomes significant, while sizing becomes significant only for high-V_{DD} and low-V_T scenarios, as we move towards high-performance regime in Fig. 5(a).

Sensitivity formulas (12)–(14), obtained from the delay and energy models from Section II, can be used to analytically calculate results from Figs. 4 and 5. Partial

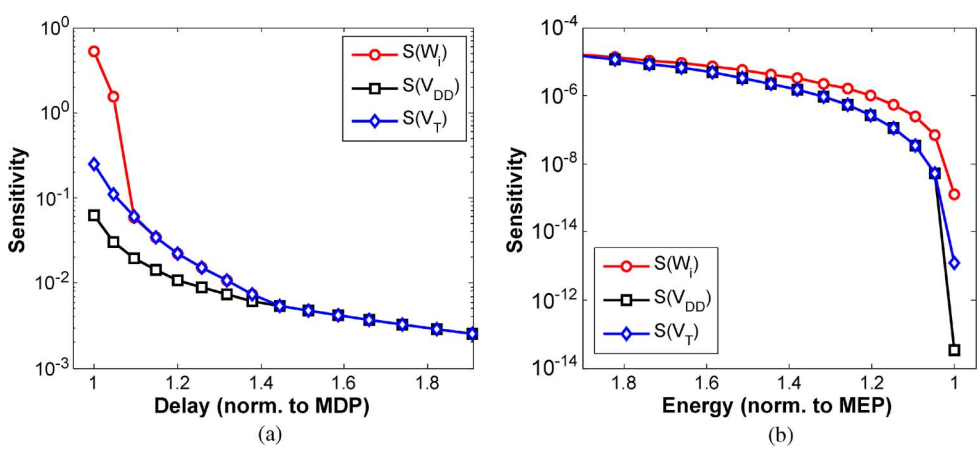

Fig. 5. ***Energy-delay sensitivity*** $S(x) = (\partial E/\partial x)/(\partial D/\partial x)$ ***near (a) MDP and (b) MEP for a 32-bit carry look-ahead adder from Fig. 4.***

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derivatives with respect to V_{DD}, V_T, and W_i lead to the following sensitivity results:

$$S_{V_{DD}} = \frac{\partial E/\partial V_{DD}}{\partial D/\partial V_{DD}} = \frac{E_{sw}}{D} \cdot \frac{2}{1-N_0} + \frac{E_{lk}}{D} \cdot \frac{2+\sigma \cdot \frac{V_{DD}}{n \cdot \phi_t} - N_0}{1-N_0}$$

$$N_0 = \frac{1+\sigma}{\sqrt{IC}} \cdot \frac{V_{DD}}{n \cdot \phi_t} \tag{12}$$

$$S_{V_T} = \frac{\partial E/\partial V_T}{\partial D/\partial V_T} = \frac{E_{lk}}{D} \cdot (1-\sqrt{IC}) \tag{13}$$

$$S_{W_i} = \frac{\partial E/\partial W_i}{\partial D/\partial W_i} = \frac{ec_i}{K_d \cdot \frac{V_{DD}}{IC} \cdot (f_{i-1} - f_i)} + \frac{E_{lk}}{D} + \frac{E_{lk}}{K_d \cdot \frac{V_{DD}}{IC} \cdot (f_{i-1} - f_i)} \tag{14}$$

where f represents the effective fanout $f = g \cdot h$ for a gate.

To demonstrate the sensitivity of V_{DD} and W_i in the E-D space, Fig. 6 plots energy-delay optimization space when V_{DD} and W_i are individually tuned, starting from MEP. As predicted, scaling V_{DD} is much more effective than using W_i around MEP, because more delay improvement is possible for a given increase in energy. Actually, sizing is hardly effective until we get close to MDP.

Therefore, unlike MDP where sizing was the most dominant optimization variable, supply voltage should be used around MEP. This is because at MEP leakage current/energy is linear function of W_i and so is performance, while V_{DD} is more effective for performance increase than sizing because V_{DD} exponentially affects performance. Given the large disparity in sizing and supply sensitivities, we may reduce sizing (if possible) around MEP to create energy slack that can be utilized by a small increase in V_{DD} for overall performance increase. This is similar, albeit in different order of adjusting variables, to increasing V_{DD} around MDP to create timing slack that can be utilized by sizing for overall energy reduction [1]. These trade-offs are generally not possible at MEP/MDP since the sizing and supply variables reach their bounds at these extreme points, so the use of sizing (MDP) or V_{DD} (MEP) is the most optimal. Indeed, this is really good news for MEP region, because supply adjustment is easier to do than to adjust gate sizing. Gate sizing involves many more variables than simple V_{DD} scaling. Besides, global V_{DD} scaling does not require any layout changes and could be done after chip fabrication.

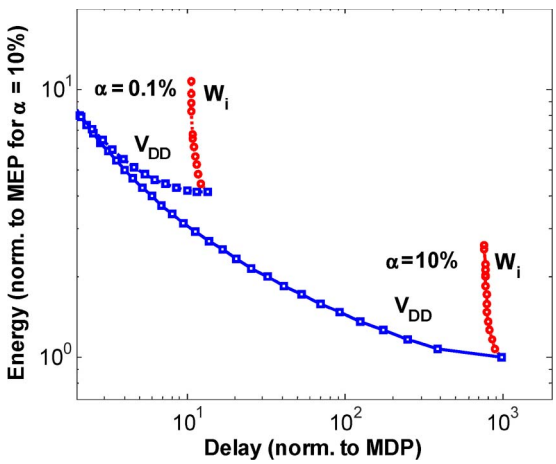

Fig. 6. ***Energy-delay trade-off after gate sizing (W_i) and voltage scaling (V_{DD}) for different activity levels for a 32-bit carry look-ahead adder from Fig. 4.***

IV. ENERGY-DELAY OPTIMIZATION

Most practical systems involve supply and sizing optimization, while threshold is selected from the available discrete values. This section explores supply and sizing optimizations for low- and high-V_T devices to compare options offered by the two thresholds. The optimization will then be expanded to include V_T, which can be performed at the device level (e.g., body-bias) and at the circuit level (e.g., type of logic family).

We start the optimization from MEP as a reference. Unlike MDP, which is a fixed point in the E-D space, MEP depends on circuit activity. Let's then first examine MEP as a function of activity factor and V_T. The discussion below is based on the 32-bit carry look-ahead adder example.

Plots in Fig. 7 show MEP and IC versus activity for high- and low-V_T designs. Since MEP is leakage-limited, HVT will always yield lower energy at the same activity. Under a very low activity factor, total energy of the circuit is dominated by its leakage energy, therefore the high-V_T cells gain significant advantage for low activity factors. For activity factor of 0.01%, for example, MEP of the HVT design achieves a 10-times lower leakage energy compared to the LVT design. Even under a high-activity factor of 10%, MEP of the HVT design is still lower in energy than that of the LVT design. It is also interesting to observe that IC corresponding to MEP greatly varies with the activity factor. For $\alpha = 0.1\%$, $IC = 5$ minimizes energy for low-V_T devices, while for $\alpha = 10\%$, MEP occurs around $IC = 0.03$ [Fig. 7(b)]. MEP is important, because it is the starting point in our optimizations. The plots in Fig. 7 do not indicate performance, which must be considered for a complete E-D comparison.

Optimal energy-performance trade-off of the same adder is shown in Fig. 8, along with the corresponding IC and V_{DD} curves in Fig. 9. From the E-D plot in Fig. 8, it is evident that although high-V_T cells achieve lower energy-per-operation than low-V_T cells, HVT has 10- to 100-times lower performance than LVT. Such large performance penalty for marginal energy reduction is highly undesirable in ULP design. For performance-constrained low-power

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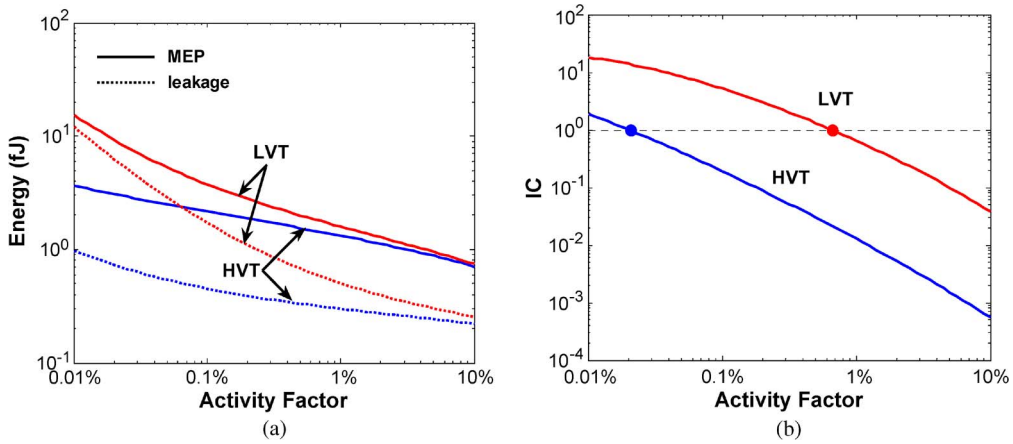

Fig. 7. ***Plot of (a) MEP and leakage energy, and (b) IC corresponding to MEP vs. activity factor for high-V_T and low-V_T devices used to implement a 32-bit carry look-ahead adder.***

designs, it is generally more effective to use low-V_T cells and operate at a lower V_{DD} than to use high-V_T cells that also require higher V_{DD} to meet the performance.

Since delay and V_{DD} depend on V_T and not on the activity factor, each delay point on the E-D curve maps to one V_{DD} point. Fig. 9(b) shows that near-threshold operation corresponds to V_{DD} of about 0.5 V for our 65-nm process. The actual V_{DD} value that minimizes energy depends on performance and the activity factor.

From Fig. 8, it is evident that low activity factors are less energy efficient. This is because designs with low activity are more affected by leakage, which increases exponentially under supply voltage reduction. Leakage could also be reduced effectively without changing the activity by reducing area via time-multiplexing. To obtain higher energy efficiency, therefore, it is beneficial to perform architectural optimizations to lower the impact of leakage, as will be discussed in Section VI. For high-performance designs, however, energy is dominated by switching energy, so activity factor plays a less significant role.

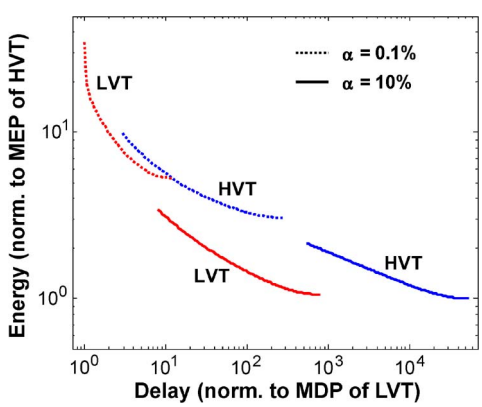

Fig. 8. ***Energy vs. delay for a 32-bit carry look-ahead adder under different activity factors α and threshold voltage V_T.***

Since high-V_T and low-V_T cells are each optimal for different regions of the energy-delay space, it is interesting to observe the outcome if we allow V_T as an optimization parameter. The optimization results of the same adder are shown in Figs. 10 and 11 where V_T is allowed to vary.

Given the freedom to vary V_T, we can achieve equal or better energy-delay efficiency than any fixed-V_T cells across all regions of the energy-delay space, for all activity factors. For the lower activity factor of 0.1%, varying V_T achieves similar energy efficiency as the HVT cells for very low-power regimes while preserving the advantage of LVT cells in the high-performance region. For the higher activity factor of 10%, any of the two fixed V_T's is suboptimal except for a very narrow range of performance. Varying V_T gives a clear advantage in the energy-delay space. It is thus desired to effectively vary V_T for improved energy-delay efficiency for an arbitrary activity. This important optimization variable is often difficult to implement in traditional CMOS processes (body bias is most commonly used). We will show that V_T control can be achieved by changing logic gate topology and circuit architecture. The SAPTL design discussed in the next section demonstrates an effective approach to exploit the use of threshold voltage through logic gate design.

V. SENSE-AMPLIFIER-BASED PASS-TRANSISTOR LOGIC (SAPTL)

A common conception of operating in the above-threshold region is that this implies supply voltages at or above 0.5 V, as threshold voltages in advanced low-power technologies (at 65 nm and below) are typically situated around 0.4 V. This ultimately limits the energy savings that can be obtained. However, operation in the above-threshold region can be achieved even for very low supply voltages

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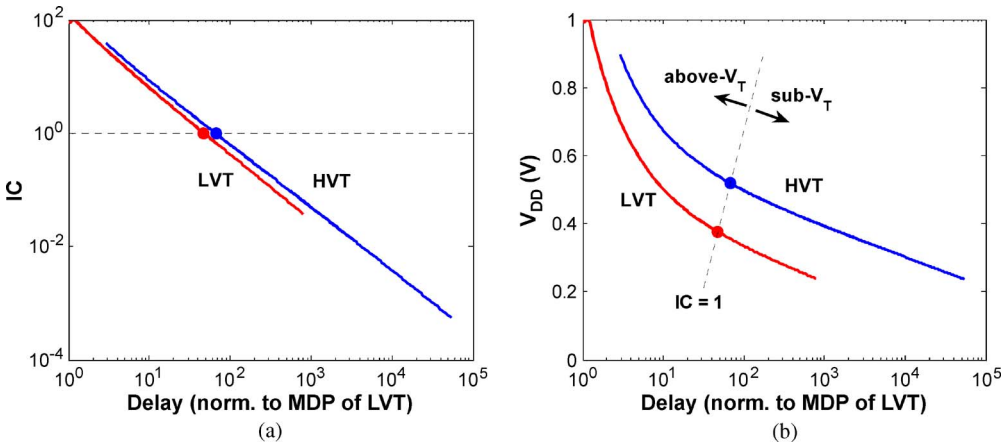

Fig. 9. ***Plot of (a) IC and (b) V_{DD} vs. delay for a 32-bit carry look-ahead adder.***

if V_T can be further lowered without increasing the subthreshold leakage current, I_{Leakage}. This is not possible in typical complementary static CMOS circuits, where I_{Leakage} is tightly coupled to V_T. But it is definitely an option in circuits without gain, such as pass transistor networks that can be designed to have no V_{DD}-to-*GND* subthreshold leakage paths.

One logic style that falls into this class is *sense-amplifier-based pass-transistor logic* (SAPTL) [14], which attempts to decouple V_T from I_{Leakage} by using pass-transistor (PT) networks to perform logic functions. The needed gain is provided using sense amplifiers and drivers, as illsutrated in Fig. 12.

The SAPTL is composed of: a) a PT network called the stack; b) a root driver; and c) a sense amplifier (SA). SAPTL can operate synchronously using a clock, or asynchronously using additional hand-shaking circuitry. The stack has a single root node energized by the driver to ensure feedforward-only operation. The function inputs steer the root current to either of the two pseudo-differential output nodes to signal either a logic "1" or a logic "0." This will produce a small voltage difference δV_{stack} at the output of the stack. This voltage is then detected and restored to full-rail by the sense amplifier.

Since the pass-transistor stack has no V_{DD} and *GND* connections, the only effect of subthreshold leakage in the pass-transistors is a deterioration of δV_{stack} at the pseudo-differential output nodes, Fig. 12. This also implies that reducing the stack threshold voltage $(V_{T,\text{stack}})$ reduces the stack delay without any subthreshold leakage penalty. The only V_{DD}-to-*GND* leakage paths appear in the sense amplifier and the driver.

This separation of concerns allows for simultaneous optimization of logic performance and static power dissipation. To maximize the logic performance, the thresholds of the pass-transistors can be lowered until

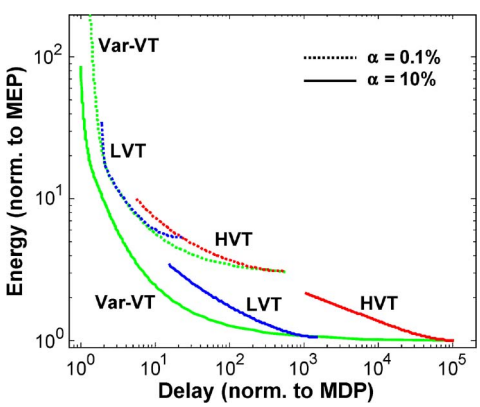

Fig. 10. ***Energy vs. delay for a 32-bit carry look-ahead adder for various V_T adjustment options (LVT, HVT, variable V_T).***

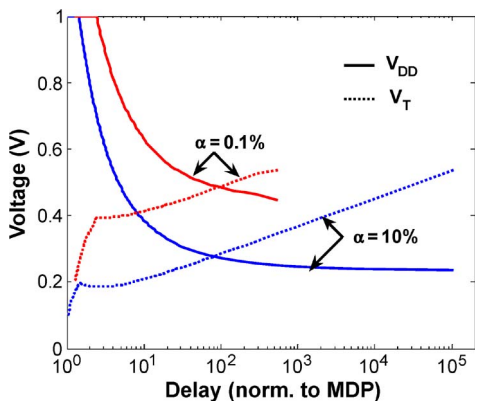

Fig. 11. ***Optimal supply and threshold voltage vs. delay for a 32-bit carry look-ahead adder after sizing, supply and threshold voltage optimizations in Fig. 10. Lower activity dictates higher voltage.***

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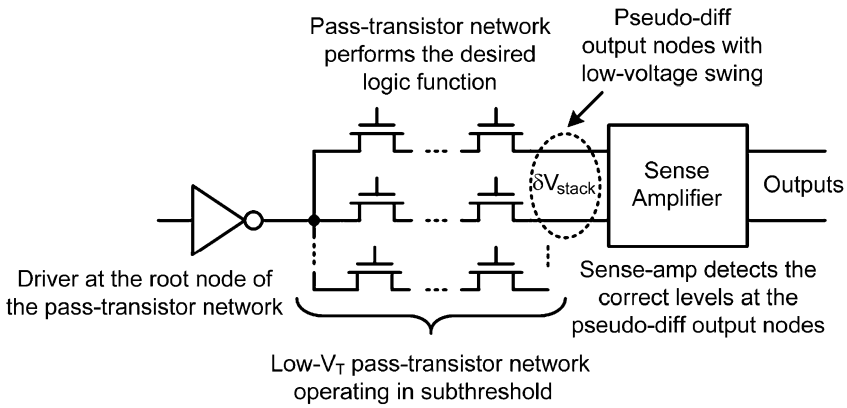

Fig. 12. ***Sense-amplifier-based pass-transistor logic (SAPTL) basic architecture.***

the energy needed by the sense amplifier to resolve the correct stack output becomes too large. Typically, a $\delta V_{\text{stack}} > 100$ mV is easily achievable at $V_{T,\text{stack}} \approx 100$ mV and can be detected with reasonable sense-amplifier energies, allowing the pass-transistors to operate comfortably in the $V_{T,\text{stack}} + \Delta V$ region.

Since $V_{T,\text{stack}}$ is different from the sense amplifier and driver threshold voltages, where leakage dominates at very low energy levels, operation in the near- or below-V_T region is desirable. One possible relation between threshold and supply voltages for the different components of the SAPTL is illustrated in Fig. 13. The pass-transistor stack has a threshold voltage $V_{T,\text{stack}}$ below the nominal V_T of logic. Stacking is the key factor for leakage control thus allowing for this configuration of logic gates.

The SAPTL delay can be expressed as the sum of the sense amplifier and driver delays, D_{active}, and the stack delay, D_{stack}. Assuming a simple dominant-pole model for the pass-transistor network, D_{stack} can be expressed as:

$$D_{\text{stack}} = \frac{k_1 \cdot n_{\text{depth}}^2}{V_{DD} - V_{T,\text{stack}}} \tag{15}$$

where n_{depth} is the depth of the pass-transistor network, i.e., the number of transistors traversed by the signal injected from the root to the output, and k_1 is a constant.

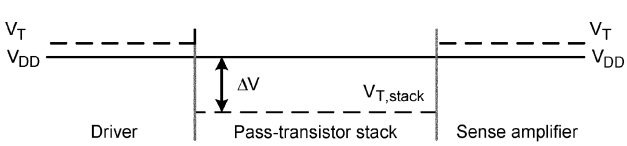

Fig. 13. ***One possible SAPTL supply and threshold voltage scenario showing subthreshold operation in the sense amplifier and driver and above-threshold operation in the pass-transistor stack.***

Thus, we can express the total SAPTL delay over M identical stages as:

$$D_{\text{SAPTL}} = M \cdot D_{\text{active}} + M \cdot \frac{k_1 \cdot n_{\text{depth}}^2}{V_{DD} - V_{T,\text{stack}}}. \tag{16}$$

Note that if the delay of the stack dominates, then reducing $V_{T,\text{stack}}$ is an effective way of reducing the delay.

The energy required by the SAPTL for a single operation is thus:

$$E_{\text{SAPTL}} = M \cdot C \cdot V_{DD}^2 + M \cdot V_{DD} \cdot \sum_{i=1}^{n_{\text{depth}}} V_i \cdot C_i + V_{DD} \cdot I_{\text{leak}} \cdot M^2 \cdot \left(D_{\text{active}} + \frac{k_1 \cdot n_{\text{depth}}^2}{V_{DD} - V_{T,\text{stack}}} \right). \tag{17}$$

The first two terms of (17) represent the active energy used by the sense amplifier and driver. Note that the voltage swing of the internal stack nodes can be kept well below V_{DD}. The last term represents the leakage energy due to both the driver and sense amplifier.

From (17), we can see that as $V_{T,\text{stack}}$ is reduced, the leakage energy is also reduced. In practice however, this increases the current flow in the off-path stack capacitances, and thus leads to a corresponding increase in off-path node voltages, which tends to cancel-out any energy reduction, but still allowing delay improvement. If we assume that for a certain logical operation, $n_{\text{depth}} \cdot M$ is a constant, i.e., it can be implemented using either many shallow SAPTL stacks or very few but deep stacks, we can then see that stack complexity and gain can be traded off against each other to achieve a desired energy-delay operating point.

In order to understand how various logic functions are implemented, consider the pass-transistor stack that implements a 4-input XOR function as shown in Fig. 14.

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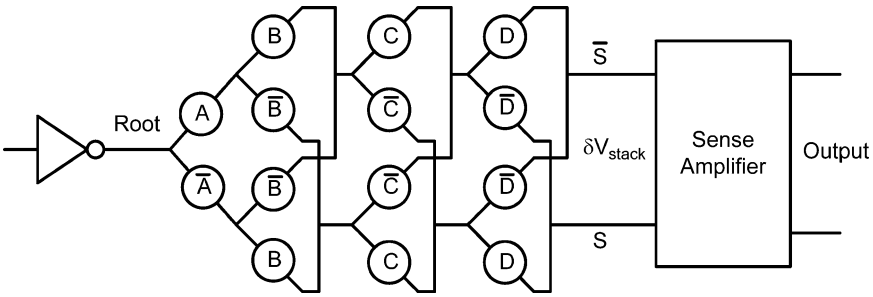

Fig. 14. ***A 4-input SAPTL XOR showing the pass transistor stack structure where each circle represents an NMOS transistor controlled by the corresponding input variable.***

Each path from the root of the stack to S represents a minterm and each path from the root to $\bar{S}$ represents a maxterm. It can be observed from Fig. 14 that the SAPTL implementation of XOR gates is very straightforward. By increasing the complexity of the stack, in this case increasing the number of inputs to the XOR gate, the sense amplifier and driver overhead per input can be reduced, at the expense of decreased performance. This can be seen in Fig. 15, where the energy and delay of a 6-input and 16-input SAPTL XOR gate are compared to their static CMOS equivalents. With the same V_T (equal to low-V_T), SAPTL reduces energy below MEP of CMOS due to longer stacks (higher effective V_T) and lower leakage.

The capability of SAPTL to decouple I_{Leakage} and $V_{T,\text{stack}}$ is illustrated using a self-timed 64-byte parallel CRC16 generator (as used in error detection). The threshold voltages of the pass-transistors (implemented using low-V_T devices) are reduced using varying degrees of forward body biasing. The simulated results are shown in Fig. 16 with supply voltage and activity as independent parameters. The simulation results show that the overall circuit delay can be reduced with almost no impact on energy even at low activity factors such as $\alpha = 1\%$. These results are constrained by the limited effectiveness of body biasing as a means to control $V_{T,\text{stack}}$. The availability of devices with even lower threshold would be desirable as it would increase the effectiveness of SAPTL for energy reduction.

As can be seen in Fig. 16, the performance improvement through body biasing is more prominent at the higher supply voltage ($V_{DD} = 0.5\ \text{V} > V_T$), at which the delay of the stack dominates the total delay. At lower supply voltages ($V_{DD} = 0.3\ \text{V} \approx V_T$), the delay of the sense amplifier as well as the hand-shaking circuitry [15] dominates since it is near the edge of subthreshold operation, limiting the performance gains obtainable through reduction of the $V_{T,\text{stack}}$. Circuit- and logic-level techniques are foundation for architecture-level optimizations, which will be next discussed in Section VI.

VI. ARCHITECTURAL OPTIMIZATION

Just as parallelism showed to be effective for energy reduction around MDP, time-multiplexing is best suited for performance increase around MEP. Architectural

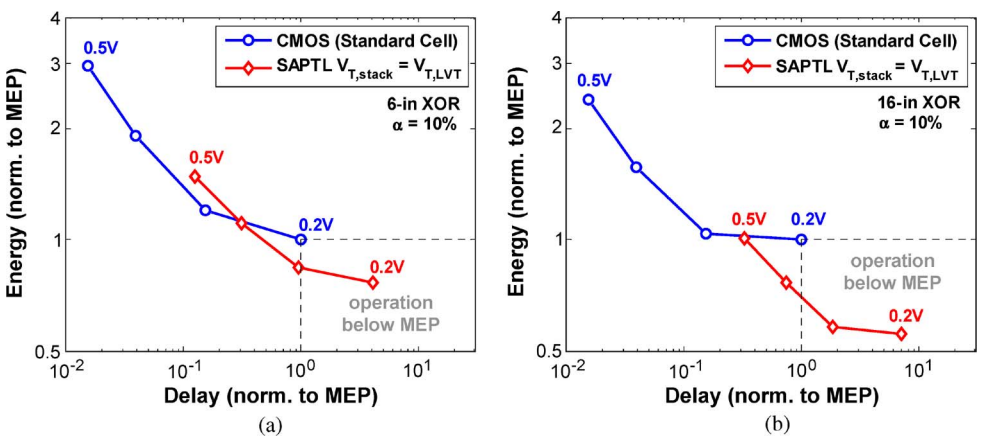

Fig. 15. ***Energy-delay characteristics of SAPTL designs: (a) 6-input XOR, (b) 16-input XOR. The plots show operation below MEP of static CMOS designs.***

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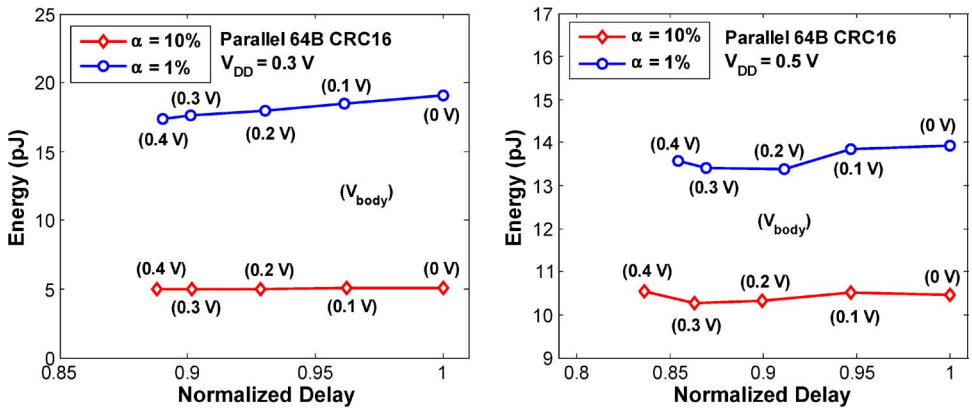

Fig. 16. ***Simulated energy-delay plots of an SAPTL 64-byte CRC16 circuit. Parameters are the body-bias of the stack transistors, supply voltage, and activity α. A forward body bias of 400 mV corresponds roughly to a 50 mV reduction in the stack threshold voltage thereby improving performance.***

optimization for improved area-performance around MEP leverages the use of time-multiplexing to even reduce energy via leakage reduction. The results from Section IV showed that a 10- to 100-times delay improvement can be gained with less than 2-times increase in the overall circuit energy. This energy-delay trade-off near MEP is very attractive for performance increase and area reduction, where parallel data streams can be processed sequentially via time-multiplexing. A time-multiplexing example is illustrated in Fig. 17(a), where two parallel datapaths are converted into one shared datapath operating at 2-times the frequency. The overhead in the mux and de-mux logic is quite small compared to the size of the logic block A. Time-multiplexing can also be applied to sequential data processing, as shown in Fig. 17(b).

By converting N parallel datapaths or folding N sequential datapaths into a single time-multiplexed datapath effectively reduces the area roughly by N times, which results in N-times reduction in leakage energy. Consequently, the delay constraint is made N-times shorter, but near the MEP point this does not result in much increase in energy, and the reduction in leakage may actually lead to decrease in total energy in leakage-dominated designs. In cases where tightening the delay constraints results in significant increase in energy, pipelining should be implemented to shorten the critical path and relax the timing penalties caused by time-multiplexing. In cases of feedback time-multiplexing, pipelining would require sequence converters to sustain original logic functions. In addition, an N-stage feedback time-multiplexed design can only sustain up to N stages of pipelining, though most designs do not benefit from excessive pipelines due to register overhead. A 2-stage pipelining scenario is shown in Fig. 18 as an example.

Fig. 19 shows energy vs. cycle time for three designs: reference, time-multiplexed, and time-multiplexed with an

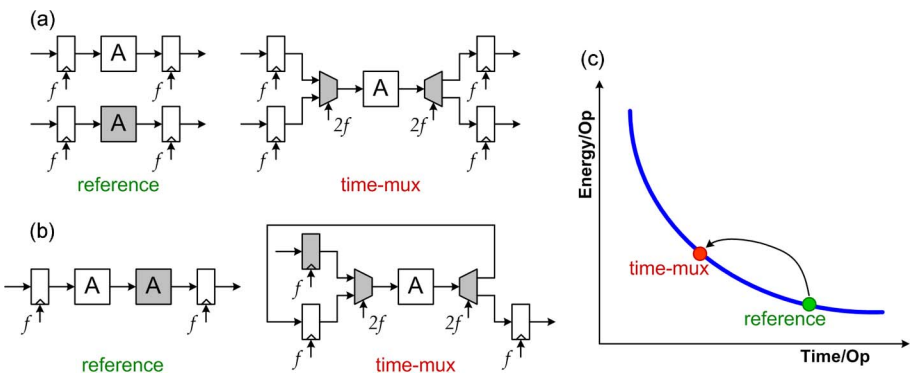

Fig. 17. ***Time-multiplexing technique applied to (a) parallel and (b) sequential data processing. Logic block A needs to run faster in the time-multiplexed designs as indicated in energy-delay space (c) of block A.***

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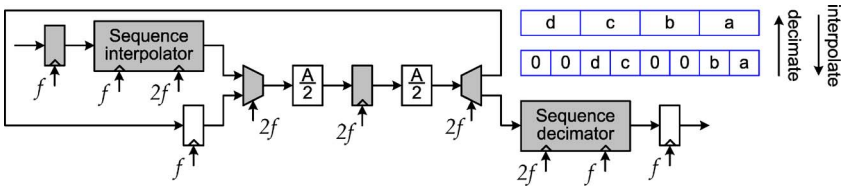

Fig. 18. ***Pipelining for feedback time-multiplexed logic.***

extra pipelining step. Numbers in brackets indicate relative throughput. Starting from the reference design (with 1× throughput), we can apply time-multiplexing to save area and also energy (due to lower leakage) while keeping the same performance. Adding pipelining may be beneficial for certain throughput targets. Although pipelining shortens the critical path, a potential energy reduction due to the delay slack has to be weighed against the energy overhead due to pipeline registers. The plot shows that for the low-throughput (0.5×) scenario near MEP, increased delay slack does not lead to significant energy reduction due to low energy-delay sensitivities, and pipelining actually causes higher energy than the time-multiplexed case due to register overhead. In most cases, however, pipelining results in reduced energy-per-operation due to a relaxed critical path.

The designs with 2 to 8 levels of time-multiplexing are shown against the reference design in Fig. 20. Each additional level of time-multiplexing lowers the leakage energy in proportion to area savings. The reduction in leakage energy results in a better energy-delay trade-off, though the reduction diminishes with each additional level of time-multiplexing. This is analogous to parallelism which reduces overall energy primarily by affecting switching energy. The additional multiplexing requires a 2-times increase in operating frequency to sustain the throughput, and since energy penalty increases with increased performance, time-multiplexing above 2 results in increased energy-per-operation. Such energy increase can be alleviated by pipelining, which results in a significant energy reduction for performance-critical designs. Due to a higher energy-delay sensitivity of these high-performance designs, pipelining achieves larger energy reduction when combined with a higher order time-multiplexing.

The initial area reduction due to time-multiplexing is the most significant; each additional level provides incrementally smaller reduction in area since the number of registers stays constant. For a reference design with roughly 10% register area, its 8-way time-multiplexed design would have a register area of about 50%. Pipelining also results in additional area. For designs with a low-order time-multiplexing, the small energy reduction (if any) due to pipelining may not justify the area penalty of the pipeline registers. The area penalty is larger for feedback time-multiplexed designs, where pipelining requires additional overhead in data-sequence converters.

Architectural techniques heavily rely on V_{DD} scaling, which is not regularly supported by chip synthesis tools. The use of ULP techniques within synthesis environment will be discussed in the next section.

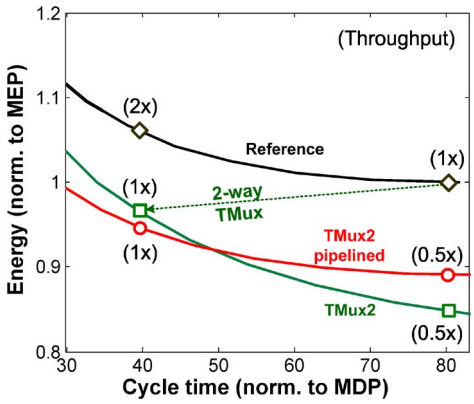

Fig. 19. ***Energy-per-operation as a function of cycle time for reference, time-multiplexed, and time-multiplexed with pipelining designs. At a reference throughput (1×), time-multiplexing saves both area and energy due to lower leakage. The energy benefits of additional pipelining are dependent on throughput: energy is lowered for 1×, increased for 0.5× throughput.***

VII. CHIP SYNTHESIS, DESIGN EXAMPLE

A. Chip Synthesis for ULP

Applying the ultralow-power design techniques within commercial synthesis tool-flows is challenging, because digital libraries are typically characterized for a fixed voltage. We cannot directly specify reduced voltage requirement during chip synthesis. Gate-size optimization within synthesis tools is also somewhat constraining, because we cannot precisely control sizing of the gates. Our control over sizing is via timing constraints and the use of various compilation techniques available. Within the constraints of synthesis tools, we can implement the

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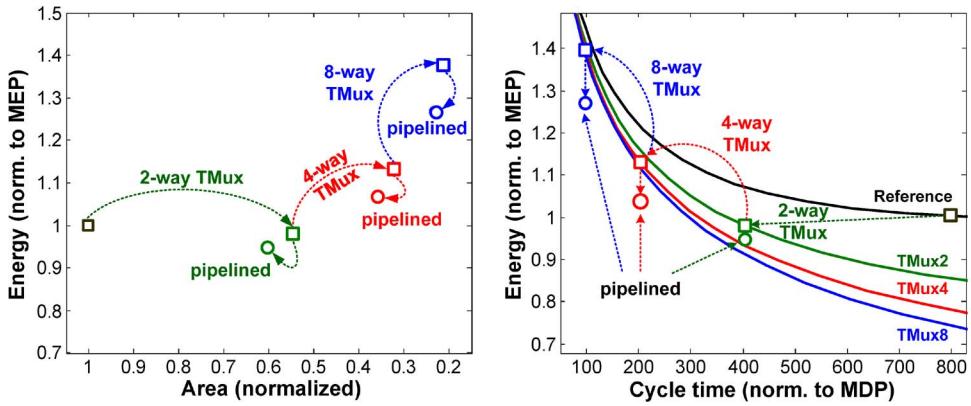

Fig. 20. ***Energy-delay and energy-area plots for designs with 2 to 8 levels of time-multiplexing for a 1× throughput as defined in Fig. 19. The results of additional pipelining are also shown. Time-multiplexing by 2 improves both energy and area, while higher levels of time-multiplexing only improve area and increase energy.***

ULP design techniques using simple heuristicsfor gate size and supply voltage adjustment.

The main ULP technique is supply voltage scaling. To design for reduced V_{DD}, we need to indirectly specify the timing constraint to chip synthesis tools that work with nominal V_{DD}. To estimate timing at nominal V_{DD}, we perform transistor-level simulations of a chain of logic gates (could be simply a FO4 inverter) to obtain delay vs. V_{DD} plot. From this plot, assuming that path delay scales the same way, we can calculate how much faster a design will have to be when operating at nominal supply voltage. This method is robust, because V_{DD} can be re-adjusted after chip fabrication, if needed, to a desired performance. Performing chip synthesis at a very high clock rate may violate timing of digital I/O cells. A way around that is to perform synthesis without I/O cells, but keep all primary inputs and outputs registered for timing closure.

Gate sizing is another important technique to use during synthesis. From explorations around MDP, we know that sizing benefits get utilized around 20% delay slack as compared to MDP. Around MEP, which is a region of interest, gates are nearly minimum-sized. To implement this in synthesis environment, we initially synthesize the design with a 20% delay slack. During synthesis, circuit topology is fixed. Then, a 20% longer delay is specified and incremental compilation step is applied to downsize the gates. The incremental compilation adjusts sizing of the gates without affecting the circuit topology. This way, we have a design with nearly MEP sizing, and with enough timing slack (starting from nominal V_{DD}) for voltage scaling.

B. Design Example: SVD Processor

The methodology described above can be easily applied to simple designs like multipliers, adders, FFT units, etc. where a designer can observe synthesis results and trace down all the way to individual gates to verify the results. The optimization of supply voltage and gate sizing can be hierarchically extended to arbitrary level of complexity. As a demonstration, we show the result of optimizing a complex digital signal processing (DSP) algorithm for channel decoupling in multi-antenna communications.

The algorithm [16] implements adaptive LMS-based tracking of eigen-values of a channel matrix in multi-antenna (MIMO) wireless channels. The eigen-values represent the gains of spatial sub-channels. For each antenna, eigen-modes are calculated, followed by deflation operation for the next antenna, until the algorithm is complete for all antennas, Fig. 21. The algorithm performs complex arithmetic and consists of about 400 multipliers and 300 adders, which is a much larger complexity than FFT or Viterbi algorithm. In the adaptive LMS, step size μ_i is also adaptively adjusted based on estimated channel gain (eigen-value) σ_i. Division and square root operations shown in Fig. 21 are also implemented iteratively using Newton–Rhapson formulas [33]. The iterative approach was possible due to flat-fading channel conditions. Overall, the design has 6 nested feedback loops, which make

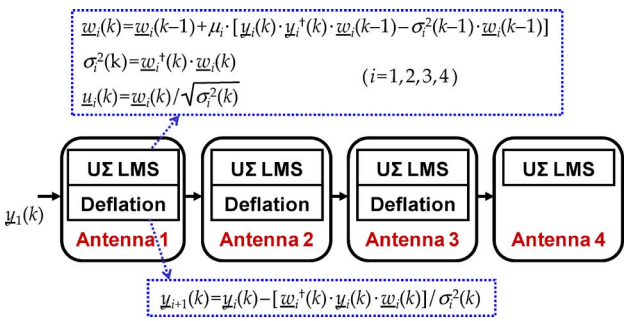

Fig. 21. ***Adaptive blind-tracking multiantenna (MIMO) algorithm for channel decoupling.***

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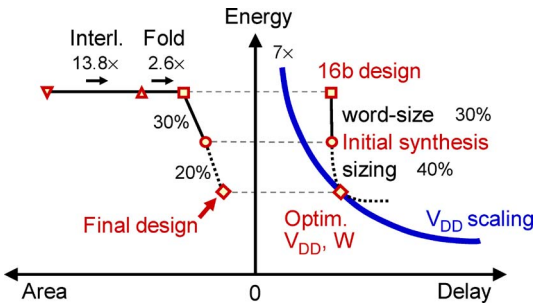

Fig. 22. ***Area-energy-delay optimization of the SVD algorithm from Fig. 21.***

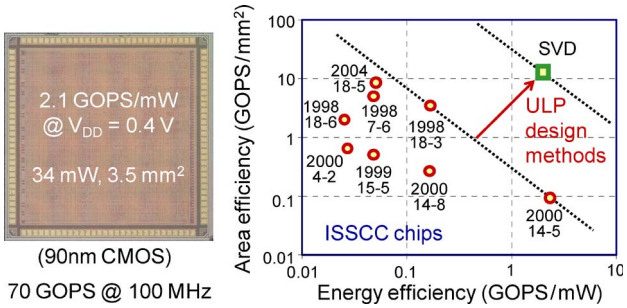

Fig. 23. ***SVD chip micrograph, its energy- and area-efficiency compared to ISSCC chips from multimedia and DSP baseband processors category. The results are normalized to a 90 nm technology. ULP design methods show clear advantage in energy- and area-cost.***

top-level retiming difficult during chip synthesis. Our approach was to characterize all blocks in the latency-cycle time space and assign good initial values for latency, so that no registers need to be pushed across the loops [17]. We then only need to do retiming of simple feedforward blocks and perform boundary optimizations during logic synthesis for hierarchical designs.

By partitioning the design as described, low-level retiming ensured balanced logic depth, which enables globally optimal supply voltage scaling. Gate sizing is done using incremental compile step in the logic synthesis. Fig. 22 shows the evolution of our design in area-energy-delay space for a variety of optimization techniques applied [18]. Starting from a 16-bit direct-mapped design, we apply wordlength optimization [19]–[21], a 4-way architecture folding for 4 antennas, a 16-way data-stream interleaving for 16 frequency subcarriers to save area and energy at the arithmetic and architectural levels. Next, we make use of gate sizing and supply voltage reduction as described in Section VII-A.

Initial synthesis is done with a 20% timing slack, after which incremental compile was performed to reduce energy via gate sizing. For 64 MHz real-time operation, timing of 512 MHz (SS corner, 0.9 V, 125 C) was specified in the logic synthesis. After supply voltage scaling, the goal is to have trade-offs with respect to sizing and V_{DD} scaling balanced (V_{DD} and W plots are tangent at the optimal point as shown in Fig. 22).

The result of our optimization is validated on a chip, Fig. 23. The chip operates at 0.4 V in a 90-nm CMOS technology with a clock rate of 100 MHz. This level of performance corresponds to operating condition close to TT and towards FF corner. The chip was designed for SS corner, hence the actual silicon was faster. The chip executes 70 GOPS (12-bit add equivalent), dissipates 34 mW of power and occupies 3.5 mm^2 of core area. This gives us energy efficiency of 2.1 GOPS/mW and area efficiency of 20 GOPS/mm^2 [22] as a result of ULP design methodology.

Key techniques for improving energy efficiency are, not surprisingly, gate sizing (40%) and supply voltage scaling (7 times). This makes sense, because energy efficiency is the amount of energy required for some average operation (12-bit add in this case). Therefore, to maximize energy efficiency, we have to minimize supply voltage and gate capacitance. This, however, has to be done in such a way to also minimize hardware area and meet the performance. Architectural techniques we used (folding, interleaving) resulted in shared datapath logic, hence greatly reducing the area for multiantenna multicarrier design. Energy and area efficiency, normalized to the same technology, of our design in comparison with similar chips [23]–[30] from the ISSCC conference is shown in Fig. 23. The use of ultralow-power techniques in chip design is clearly beneficial for energy and area minimization.

VIII. CONCLUSION

Computations in standard CMOS logic are limited by subthreshold leakage. Minimum-energy point is very expensive in terms of performance, just like minimum-delay point is in terms of energy. We can gain 10-times in performance by increasing energy by ~20% above MEP. The most dominant optimization variable around MEP is the supply voltage, while time-multiplexing helps reduce area and leakage energy. Moderate inversion operation requires careful modeling and suggests new pass-transistor based logic style that can outperform standard CMOS in the near-threshold regime, by achieving the energy below MEP of standard CMOS. The use of time-multiplexing around MEP results in both lower area and energy without performance penalty, due to reduced leakage that comes with a lower area. Voltage scaling and gate sizing techniques around MEP can be applied within conventional chip synthesis tools (in a sequential manner) to minimize both area and energy. We have demonstrated energy efficiency of 2 GOPS/mW and area efficiency of 20 GOPS/mm^2 in a 90-nm CMOS technology using the techniques described in this paper. ■

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REFERENCES

[1] V. Stojanović, D. Marković, B. Nikolić, M. Horowitz, and R. Brodersen, "Energy-delay tradeoffs in combinational logic using gate sizing and supply voltage optimization," in *Proc. Eur. Solid-State Circuits Conf.*, 2002, pp. 211–214.

[2] R. Gonzalez, B. Gordon, and M. A. Horowitz, "Supply and threshold voltage scaling for low power CMOS," *IEEE J. Solid-State Circuits*, vol. 32, no. 8, pp. 1210–1216, Aug. 1997.

[3] V. Zyuban and P. Strenski, "Unified methodology for resolving power-performance tradeoffs at the microarchitectural and circuit levels," in *Proc. Int. Symp. Low Power Electronics and Design*, 2002, pp. 166–171.

[4] T. Kuroda *et al.*, "Variable supply-voltage scheme for low-power high-speed CMOS digital design," *IEEE J. Solid-State Circuits*, vol. 33, no. 3, pp. 454–462, Mar. 1998.

[5] K. Nose and T. Sakurai, "Optimization of V_{DD} and V_{TH} for low-power and high-speed applications," in *Proc. Asia South Pacific Design Automation Conf.*, 2000, pp. 469–474.

[6] D. Marković, V. Stojanović, B. Nikolić, M. A. Horowitz, and R. W. Brodersen, "Methods for true energy-performance optimization," *IEEE J. Solid-State Circuits*, vol. 39, no. 8, pp. 1282–1293, Aug. 2004.

[7] C. Enz, F. Krummenacher, and E. Vittoz, "An analytical MOS transistor model valid in all regions of operation and dedicated to low-voltage and low-current applications," *J. Analog IC Signal Proc.*, vol. 8, no. 1, pp. 83–114, Jul. 1995.

[8] E. A. Vittoz, "Weak inversion for ultimate low-power logic," in *Low-Power Electronics Design*, C. Piguet, Ed. Boca Raton, FL: CRC, 2005.

[9] B. Calhoun, A. Wang, and A. Chandrakasan, "Modeling and sizing for minimum energy operation in subthreshold circuits," *IEEE J. Solid-State Circuits*, vol. 40, no. 9, pp. 1778–1786, Sep. 2005.

[10] B. Zhai *et al.*, "A 2.60 pJ/inst subthreshold sensor processor for optimal energy efficiency," in *IEEE Symp. VLSI Circuits Dig. Tech. Papers*, 2006, pp. 154–155.

[11] I. Sutherland, B. Sproul, and D. Harris, *Logical Effort: Designing Fast CMOS Circuits*, 1st ed. San Francisco, CA: Morgan Kaufmann, 1999.

[12] T. Sakurai and R. Newton, "Alpha-power law MOSFET model and its applications to CMOS inverter delay and other formulas," *IEEE J. Solid-State Circuits*, vol. 25, no. 2, pp. 584–594, Apr. 1990.

[13] K. Bowman, B. Austin, J. Eble, T. Xinghai, and J. Meindl, "A physical alpha-power law MOSFET model," *IEEE J. Solid-State Circuits*, vol. 34, no. 10, pp. 1410–1414, Oct. 1999.

[14] L. Alarcón, T.-T. Liu, M. Pierson, and J. Rabaey, "Exploring very low-energy logic: A case study," *J. Low Power Electron.*, vol. 3, no. 3, pp. 223–233, Dec. 2007.

[15] T.-T. Liu, L. Alarcón, M. Pierson, and J. Rabaey, "Asynchronous computing in sense amplifier-based pass transistor logic," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 17, no. 7, pp. 883–892, Jul. 2009.

[16] A. Poon, D. Tse, and R. W. Brodersen, "An adaptive multiple-antenna transceiver for slowly flat-fading channels," *IEEE Trans. Commun.*, vol. 51, no. 13, pp. 1820–1827, Nov. 2003.

[17] D. Marković, "A Power/Area Optimal Approach to VLSI Signal Processing," Ph.D. thesis, Univ. California, Berkeley, Spring, 2006.

[18] D. Marković, B. Nikolić, and R. W. Brodersen, "Power and area minimization for multidimensional signal processing," *IEEE J. Solid-State Circuits*, vol. 42, no. 4, pp. 922–934, Apr. 2007.

[19] C. Shi and R. W. Brodersen, "Automated fixed-point data-type optimization tool for signal processing and communication systems," in *Proc. IEEE Design Automation Conf.*, Jun. 2004, pp. 478–483.

[20] C. Shi and R. W. Brodersen, "A perturbation theory on statistical quantization effects in fixed-point DSP with non-stationary inputs," in *Proc. Int. Symp. Circuits Syst.*, May 2004, pp. 373–376.

[21] C. Shi, "Floating-Point to Fixed-Point Conversion," Ph.D. thesis, Univ. California, Berkeley, Spring, 2004.

[22] D. Marković, R. W. Brodersen, and B. Nikolić, "A 70 GOPS, 34 mW multi-carrier MIMO chip in 3.5 mm^2," in *Proc. Int. Symp. VLSI Circuits*, Jun. 2006, pp. 196–197.

[23] S. Santhanam *et al.*, "A low-cost 300 MHz RISC CPU with attached media processor," in *IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers*, Feb. 1998, pp. 298–299.

[24] J. Williams *et al.*, "A 3.2 GOPS multiprocessor DSP for communication applications," in *IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers*, Feb. 2000, pp. 70–71.

[25] T. Ikenaga and T. Ogura, "A fully-parallel 1 Mb CAM LSI for real-time pixel-parallel image processing," in *IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers*, Feb. 1999, pp. 264–265.

[26] M. Wosnitza, M. Cavadini, M. Thaler, and G. Troster, "A high precision 1024-point FFT processor for 2D convolution," in *IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers*, Feb. 1998, pp. 118–119.

[27] F. Arakawa *et al.*, "An embedded processor core for consumer appliances with 2.8 GFLOPS and 36 M polygons/s FPU," in *IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers*, Feb. 2004, pp. 334–335.

[28] M. Strik *et al.*, "Heterogeneous multi-processor for the management of real-time video and graphics streams," in *IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers*, Feb. 2000, pp. 244–245.

[29] H. Igura *et al.*, "An 800 MOPS 110 mW 1.5 V parallel DSP for mobile multimedia processing," in *IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers*, Feb. 1998, pp. 292–293.

[30] P. Mosch *et al.*, "A 720 μW 50 MOPs 1 V DSP for a hearing aid chip set," in *IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers*, Feb. 2000, pp. 238–239.

[31] V. Zyuban and P. Strenski, "Unified methodology for resolving power-performance tradeoffs at the microarchitectural and circuit levels," in *Proc. Int. Symp. Low Power Electron. Design*, Aug. 2002, pp. 166–171.

[32] R. Brodersen, M. Horowitz, D. Markovic, B. Nikolic, and V. Stojanovic, "Methods for true power minimization," in *Proc. Int. Conf. Comput.-Aided Design*, Nov. 2002, pp. 35–42.

[33] C. Ramamoorthy, J. Goodman, and K. Kim, "Some properties of iterative square-rooting methods using high-speed multiplication," *IEEE Trans. Comput.*, vol. C-21, no. 8, pp. 837–847, 1972.

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